

DATA PIPELINE_03: BLOB ,ADLS to SQLDB (ADF USING DataFlow)

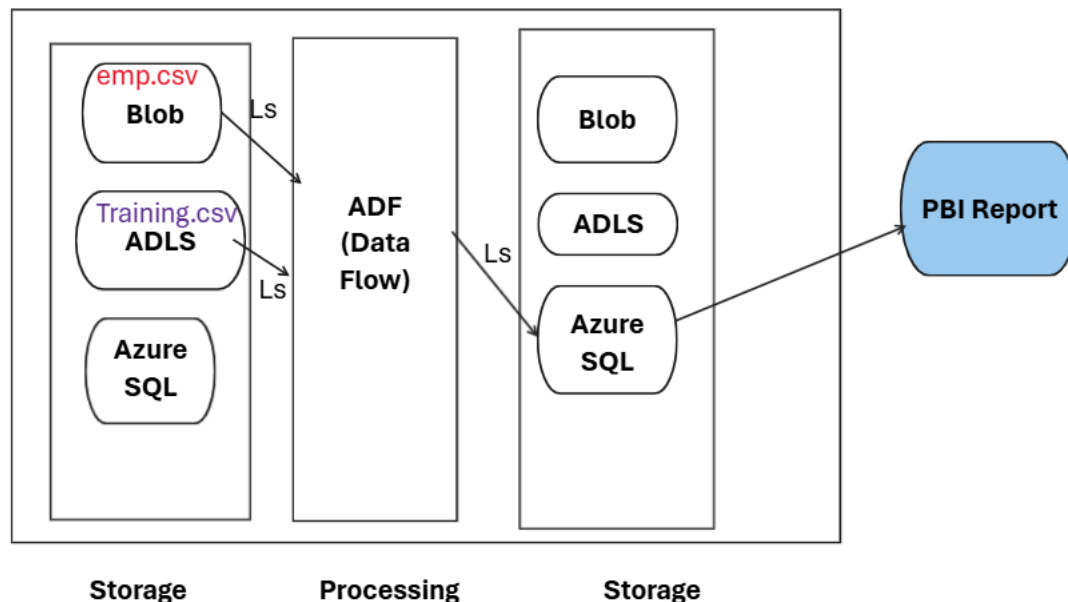
Project Objective:

To extract data from Azure Blob Storage and ADLS, transform it using Azure Data Factory Mapping Data Flows, and load it into Azure SQL Database to create a reliable, scalable, and efficient data integration solution for analytics and reporting.

Process:

Raw Data Sources (Azure Blob Storage + ADLS) → Azure Data Factory Pipeline → Mapping Data Flow (Transformations) → Azure SQL Database → Reporting/Analytics Layer

Architecture:

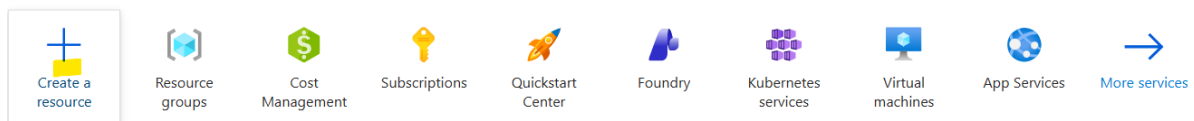


1)TO CREATE BLOB STORAGE:

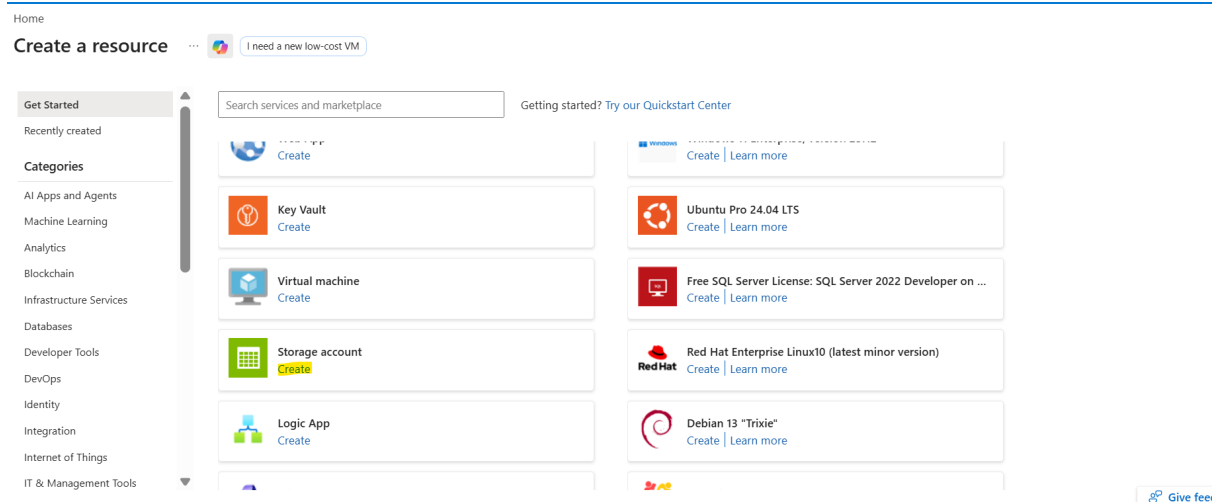
1.1) login to azure portal and signin with email id

1.2)Create Resource:

Azure services



1.3) Click on the + icon and select the storage account:



1.4) Fill all the basic details:

Subscription: Give your subscription :ex) Azure subscription1

Resource group: click create-->--new--Create a new resource group

Instance details:

Storage account name: give any name for storage account like pipelin3sa

Region: select your nearest region

Preferred storage type: Azure blob storage and azure data lake storage gen2

Performance: Standard

Redundancy: LRS depends on the data

Create a storage account ...

Subscription *	NK
Resource group *	(New) pipeline3rg Create new
Instance details	
Storage account name * ⓘ	pipeline3sa
Region * ⓘ	(US) West US 3 Deploy to an Azure Extended Zone
Preferred storage type	Azure Blob Storage or Azure Data Lake Storage i This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. Learn more
Performance * ⓘ	☒ Standard: Recommended for most scenarios (general-purpose v2 account) ☐ Premium: Recommended for scenarios that require low latency.
Redundancy * ⓘ	Locally redundant storage (LRS)

Previous

Next

Review + create

Fill in the advanced details:

Blob storage:select how often we use the data depends on(HOT,COOL,COLD)

Public network access:Enable

Public network access scope:Enable all networks

Routing preference:Microsoft Network Routing

Enable soft delete:Give the no of days you can retrieve the data once it deleted

Reference:

Create a storage account

- Basics
- Advanced
- Networking
- Data protection
- Security
- Encryption
- Tags
- Review + create

[View automation template](#)

Basics

Subscription	NK
Resource group	pipeline3rg
Location	West US 3
Storage account name	pipeline3sa
Primary service	Azure Blob Storage or Azure Data Lake Storage
Performance	Standard
Replication	Locally redundant storage (LRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled

Performance	Standard
Replication	Locally redundant storage (LRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Cold
Managed Identity for SMB	Disabled

Azure Files

Require Encryption in Transit for SMB	Enabled
---------------------------------------	---------

Security

Secure transfer	Enabled
Blob anonymous access	Disabled
Allow storage account key access	Enabled
Default to Microsoft Entra authorization in the Azure portal	Disabled
Minimum TLS version	Version 1.2
Permitted scope for copy operations (Preview)	From any storage account
Enable Defender for Storage	Disabled

Networking

Public network access	Enabled
Public network access scope	Enabled from all networks
Default routing tier	Microsoft network routing

Data protection

Point-in-time restore	Disabled
Blob soft delete	Enabled
Blob retainment period in days	7
Container soft delete	Enabled
Container retainment period in days	7
Classic file share soft delete	Enabled
Classic file share retainment period in days	7
Versioning	Disabled
Blob change feed	Disabled
Version-level immutability support	Disabled

Encryption

Encryption type	Microsoft-managed keys (MMK)
Enable support for customer-managed keys	Blobs and files only
Enable infrastructure encryption	Disabled

[Previous](#)[Next](#)[Create](#)[Home](#)**pipeline3sa_1782288421299** | Overview

Deployment



Delete



Cancel



Redeploy



Download



Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment details

Resource	Type	Status	Operation de
No results.			

[Give feedback](#)[Tell us about your experience with deployment](#)

Successfully created blob storage account

[Home](#)**pipeline3sa_1782288421299** | Overview

Deployment



Delete



Cancel



Redeploy



Download



Refresh

Overview

Inputs

Outputs

Template

Your deployment is complete



Deployment name: pipeline3sa_1782288421299

Subscription: [NK](#)Resource group: [pipeline3rg](#)

Start time: 24/6/2026, 1:39:10 pm

Correlation ID: 78912773-8a7f-41a9-9028-7c0e18f295c1



Deployment details

Next steps

[Go to resource](#)[Give feedback](#)

Go to resource---storage browser---blob container(right side menu)---create a container---add container---data-upload the file

Home > pipeline3sa_1782288421299 | Overview > pipeline3sa

pipeline3sa | Storage browser Help me save costs by tiering unused blobs

Storage account

Search

- Overview
- Activity log
- Tags
- Diagnose and solve problems
- Access Control (IAM)
- Data migration
- Events
- Storage browser**
- Storage Mover
- Partner solutions
- Resource visualizer

pipeline3sa

- Favorites
- Recently viewed
- Blob containers
- Slogs
- data

View all

- Classic file shares
- Queues
- Tables

+ Add Directory ↑ Upload 🔒 Change access le

Blob containers > data

Authentication method: Access key ([Switch to Microsoft Entr](#))

Add filter

Search blobs by prefix (case-sensitive)

Showing all 1 items

Name	Last modified
Emp1.csv	24/06/2026, 13:42:37

2)TO CREATE ADLS ACCOUNT:

*****Note:

Only difference is from the blob storage (we don't enable the hierarchical name space) but in ADLS(We enable hierarchical name space)

2.1)Create Resource:

Azure services

2.2) Click on the + icon and select the storage account:

Home

Create a resource I need a new low-cost VM

Get Started

Recently created

Categories

- AI Apps and Agents
- Machine Learning
- Analytics
- Blockchain
- Infrastructure Services
- Databases
- Developer Tools
- DevOps
- Identity
- Integration
- Internet of Things
- IT & Management Tools

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Virtual machine Create

Storage account **Create**

Logic App Create

Ubuntu Pro 24.04 LTS Create | Learn more

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Red Hat Enterprise Linux10 (latest minor version) Create | Learn more

Debian 13 "Trixie" Create | Learn more

Give feedback

2.3)Fill all the basic details:

Subscription: Give your subscription :ex)Azure subscription1

Resource group: click create---new-Create a new resource group

Instance details:

Storage account name: give any name for storage account like pipelin3adlsa

Region: select your nearest region

Preferred storage type: Azure blob storage and azure data lake storage gen2

Performance: Standard

Redundancy: LRS depends on the data

Home > [Create a resource](#)

Create a storage account ...

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription *	NK
Resource group *	pipeline3rg Create new

Instance details

Storage account name * ⓘ	pipeline3adlsa
Region * ⓘ	(US) West US 3 Deploy to an Azure Extended Zone
Preferred storage type	Choose preferred storage type
Performance * ⓘ	☒ Standard: Recommended for most scenarios (general-purpose v2 account) ☐ Premium: Recommended for scenarios that require low latency.
Redundancy * ⓘ	Locally redundant storage (LRS)

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[Review + create](#)

Fill in the advanced details:

Blob storage:select how often we use the data depends on(HOT,COOL,COLD)

Public network access:Enable

Public network access scope:Enable all networks

Routing preference:Microsoft Network Routing

*****Hierarchical Namespace:Enable

Enable soft delete:Give the no of days you can retrive the data once it deleted

Create a storage account ...

- Basics
- Advanced
- Networking
- Data protection
- Security
- Encryption
- Tags
- Review + create

Hierarchical Namespace

Hierarchical namespace, complemented by Data Lake Storage Gen2 endpoint, enables file and directory semantics, accelerates big data analytics workloads, and enables access control lists (ACLs) [Learn more](#)

Enable hierarchical namespace 

[View automation template](#)

Basics

Subscription	NK
Resource group	pipeline3rg
Location	West US 3
Storage account name	pipeline3adlsa
Primary service	
Performance	Standard
Replication	Locally redundant storage (LRS)

Advanced

Enable hierarchical namespace	Enabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Cold
Managed Identity for SMB	Disabled

- Previous
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- Create

Home > Create a resource

Create a storage account

Basics Advanced Networking Data protection Security Encryption Tags Review + create

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Basics

Subscription	NK
Resource group	pipeline3rg
Location	West US 3
Storage account name	pipeline3adlsa
Primary service	
Performance	Standard
Replication	Locally redundant storage (LRS)

Advanced

Enable hierarchical namespace	Enabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Cold
Managed Identity for SMB	Disabled

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[Give feedback](#)

*** **Submitting deployment...**
Submitting the deployment template for resource 'pipeline3rg'.

Home



pipeline3adlsa_1782289224022 | Overview

Deployment

[Delete](#) [Cancel](#) [Redeploy](#) [Download](#) [Refresh](#)

Overview

[Inputs](#)

[Outputs](#)

[Template](#)

Deployment is in progress



Deployment name: pipeline3adlsa_1782289224022
Subscription: [NK](#)
Resource group: [pipeline3rg](#)

Start time: 24/1
Correlation ID:

Deployment details

Resource	Type
pipeline3adlsa	Microsoft.Storage/storageAccounts

Give feedback

[Tell us about your experience with deployment](#)

Successfully adls storage account is also created

Home



pipeline3adlsa_1782289224022 | Overview ...

Search

Delete Cancel Redeploy Download Refresh

Overview

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Template

Your deployment is complete



Deployment name: pipeline3adlsa_1782289224022
Subscription: [NK](#)
Resource group: [pipeline3rg](#)

Start time: :
Correlation

Deployment details

Next steps

[Go to resource](#)

Give feedback

[Tell us about your experience with deployment](#)

Click on Go to resource---storage browser---Blob containers---add container---we can add directory---can upload the data

Home > pipeline3adlsa_1782289224022 | Overview > pipeline3adlsa

pipeline3adlsa | Storage browser ... [Help me save costs by tiering unused blobs](#)

Search

Add Directory Upload Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

Blob containers > practice > dataset

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	[]					
☐	training11111	24/06/2026, 13:54:48	Cold (inferred)	Block blob	25.73 KiB	Available

3)TO CREATE AZURE SQL DATABASE

Azure services



Click+---to create resource---select the database and create

Home

Create a resource

...  I need a new low-cost VM

Get Started

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Categories

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SQL Database
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Azure SQL
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Azure Cosmos DB
[Create](#)



Azure Database for PostgreSQL Flexible Server
[Create](#)



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PostgreSQL on Ubuntu 24.04
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Percona Server for MySQL 5.7 on Ubuntu
[Create](#) | [Learn more](#)

Fill in all the basic details:

- Click "Create a resource" (top left)
- Search for "SQL Database", then click Create
- Fill in the basics:
 - o Subscription: Choose your subscription
- Resource Group: Select the one you created
- Database name: Give database name
- Server: Click "Create new"
- **Create a SQL Server :**
 - This is where your database will present.
 - Click Create new (under SQL Server)
- Server name: (must be unique)
- Location: Same as resource group
- Admin login: Create a username
- Password: Set a strong password o Click OK to return to the SQL DB creation form.
- **Configure Database(DTU)Basic**
- **Enable Public Access (Firewall Rule):**
 - Connectivity Method: Public Endpoint
 - In the Networking tab, under Firewall rules, select: Allow Azure services and resources to access this server: yes
- Click Review + create
- Wait for validation
- Click Create

Create SQL Database ...

Microsoft

- Basics
- Networking
- Security
- Additional settings
- Tags
- Review + create

Product details

SQL database
by Microsoft
[Terms of use](#) | [Privacy policy](#)

Estimated cost per month
4.90 USD

Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact transactional information with the provider(s) of the offering(s) for support, billing and other transactional Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace](#)

Basics

Subscription	NK
Resource group	pipeline3rg
Region	West US 2
Database name	pipeline3db
Server	(new) pipeline3server
Authentication method	SQL authentication

Create

< Previous

[Download a template for automation](#)

Home

Microsoft.SQLDatabase.newDatabaseNewServer_3b9f0fc844644a4aa6b3a | Overview ...

Deployment

Search Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServer_3b9f0fc844644a4aa6b3a Start time : 24/6/2026, 2:04:26 pm
Subscription : NK Correlation ID : 9ceab9d7-b6e1-4809-8e6e-3dc9db4f9e3c
Resource group : pipeline3rg

Deployment details

Resource	Type	Status
pipeline3server	Microsoft.Sql/servers	Accepted

Succesfully created Azure sql db

Home

Microsoft.SQLDatabase.newDatabaseNewServer_3b9f0fc844644a4aa6b3a | Overview ...

Deployment

Search x << Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServer_3b9f0fc844644a4aa6b3a Start time : 24/6/2026, 2:04:27 pm
 Subscription : NK Correlation ID : 9ceab9d7-b6e1-4809-8e6e-3dc9db4f9e3c
 Resource group : pipeline3rg

> Deployment details

✓ Next steps

[Go to resource](#)

Click on the go to resource---query editor(preview)---give sql authentication(username,password)

Home > Microsoft.SQLDatabase.newDatabaseNewServer_3b9f0fc844644a4aa6b3a | Overview > pipeline3db (pipeline3server/pipeline3db)

pipeline3db (pipeline3server/pipeline3db) | Query editor (preview) ☆ ...

SQL database

Search ○ << New query Templates Open in Feedback Sign out Classic experience

Overview

Activity log

Tags

Diagnose and solve problems

Query editor (preview)

Mirror database in Fabric (preview)

Resource visualizer

> Settings

> Data management

> Integrations

> Power Platform

> Security

> Intelligent performance

> Monitoring

> Automation

> Help

Explorer <<

Search

pipeline3db

> dbo

> Queries



Start exploring

Write and run SQL queries, create database objects, and more. [View documentation](#)

[New query](#)

4)TO CREATE ADF:

Azure services



Home

Create a resource

...



I need a new low-cost VM

Get Started

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Media

...

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MongoDB Products & Services
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Fill in the basic details required:

SubscriptionName:Give your subscription

Resource group: Pipeline3rg

Name: pipeline3adfbike

Region: select region

Version: V2

Networking:

Connect via:Public endpoint

Create Data Factory ...

- Basics
- Git configuration
- Networking
- Advanced
- Tags
- Review + create

 [View automation template](#)

Basics

Subscription	NK
Resource group	pipeline3rg
Name	pipeline3adfbike
Region	East US
Version	V2

Networking

Connect via	Public endpoint
-------------	-----------------

Previous

Next

Create

Click on create---wait for deployment

Sucessfully created ADF

C

Home



Microsoft.DataFactory-20260624141224 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : Microsoft.DataFactory-20260624141224
Subscription : NK
Resource group : pipeline3rg

> Deployment details

✓ Next steps

Go to resource

Give feedback

Tell us about your experience with deployment

Go to resource ---click on the data factory---launch azure data studio

pipeline3adfbike
Data factory (V2)

How do I troubleshoot pipeline failures in this Data factory (V2)? Summarize the properties of this Data factory (V2). List triggers associated with this Data factory (V2).

Search

Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Getting started

Monitoring

Automation

Help

Essentials

Resource group (move) : pipeline3rg
Status : Succeeded
Location : East US
Subscription (move) : NK
Subscription ID : 5bd8f1cd-03ad-48fa-8410-6e58274fc11f
Type : Data factory (V2)
Getting started : Quick start



Azure Data Factory Studio

Launch studio

Quick Starts



Tutorials

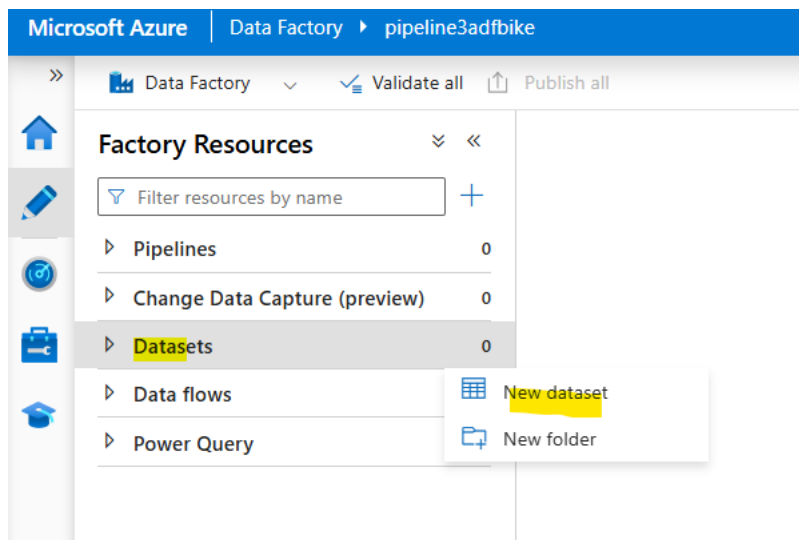


Template Gallery



Training Modules

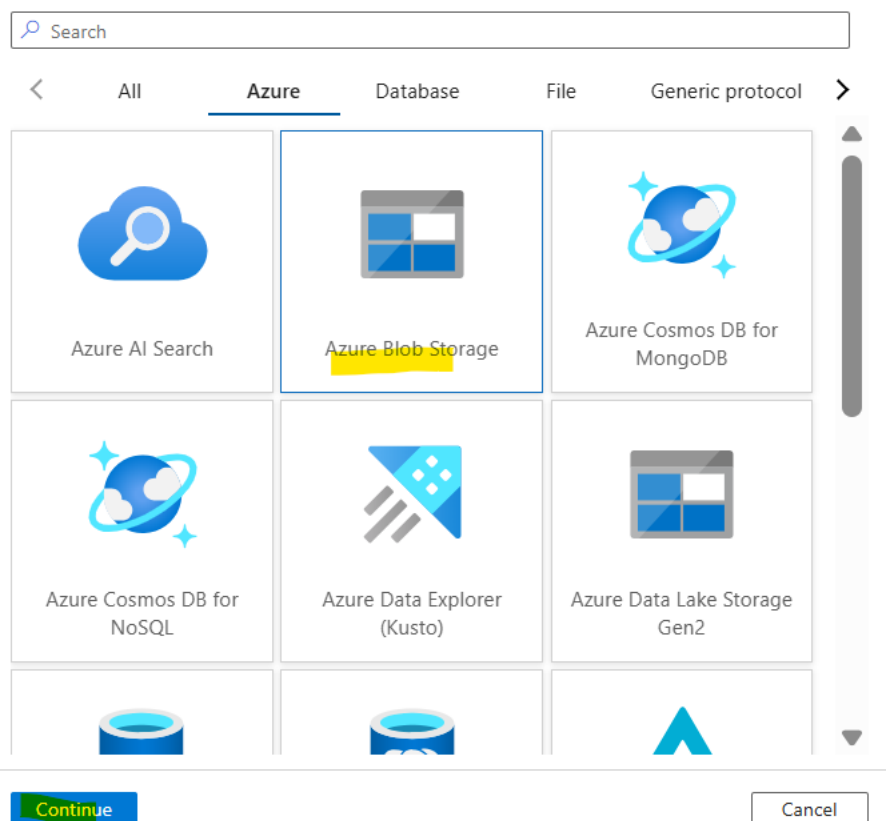
We need to copy the raw data from the storage accounts(blob and adls)



New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store



Create a new linked service to copy data from blob 2 adf---give subscription---storage account---test connection

New linked service

 Azure Blob Storage [Learn more](#) 

Authentication type

Account key

Connection string

[Azure Key Vault](#)

Account selection method ^①

☒ From Azure subscription ☐ Enter manually

Azure subscription ^①

NK (5bd8f1cd-03ad-48fa-8410-6e58274fc11f)

Storage account name *

pipeline3sa

Additional connection properties

+ New

Test connection ^①

☒ To linked service ☐ To file path

Annotations

+ New

> Parameters

Create

Cancel

✔ Connection successful

 Test connection

Now provide the file path

Set properties

Name

emp_data

Linked service *

blob2adf

File path

data

/ Directory

/ Emp1.csv

First row as header



Import schema

☒ From connection/store ☐ From sample file ☐ None

OK

Back

Cancel

Preview the data to confirm whether the data is loaded or not

Preview data

Linked service: blob2adf

Object: Emp1.csv

	EmployeeID	ManagerID	Title	MaritalStatus	Gender	HireDate	Dept	Job Grade
1	1	16	Gustavo Achong	M	M	02-02-2013 00:00	Sales	Admin
2	2	6	Catherine Abel	S	M	31-08-2013 00:00	Sales	Management
3	3	12	Kim Abercrombie	M	M	16-06-2014 00:00	Finance	Admin
4	4	3	Humberto Acevedo	S	M	10-07-2014 00:00	Logistics	Admin
5	5	263	Pilar Ackerman	M	M	16-07-2014 00:00	Human Resource	Admin
6	6	109	Frances Arams	S	M	25-07-2014	Sales	Management

Connection

Linked service *

File path *

Compression type

Column delimiter

Row delimiter

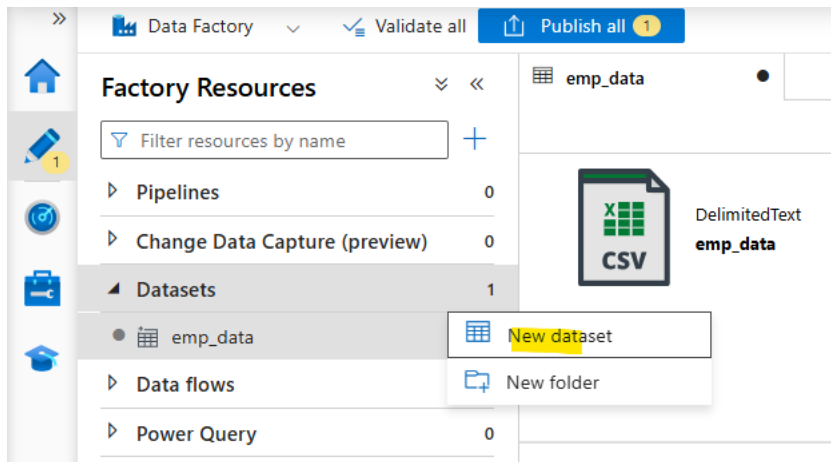
Encoding

Quote character

Escape character

Backslash (\)

***we need to create one more dataset because the data we are getting from 2 different sources

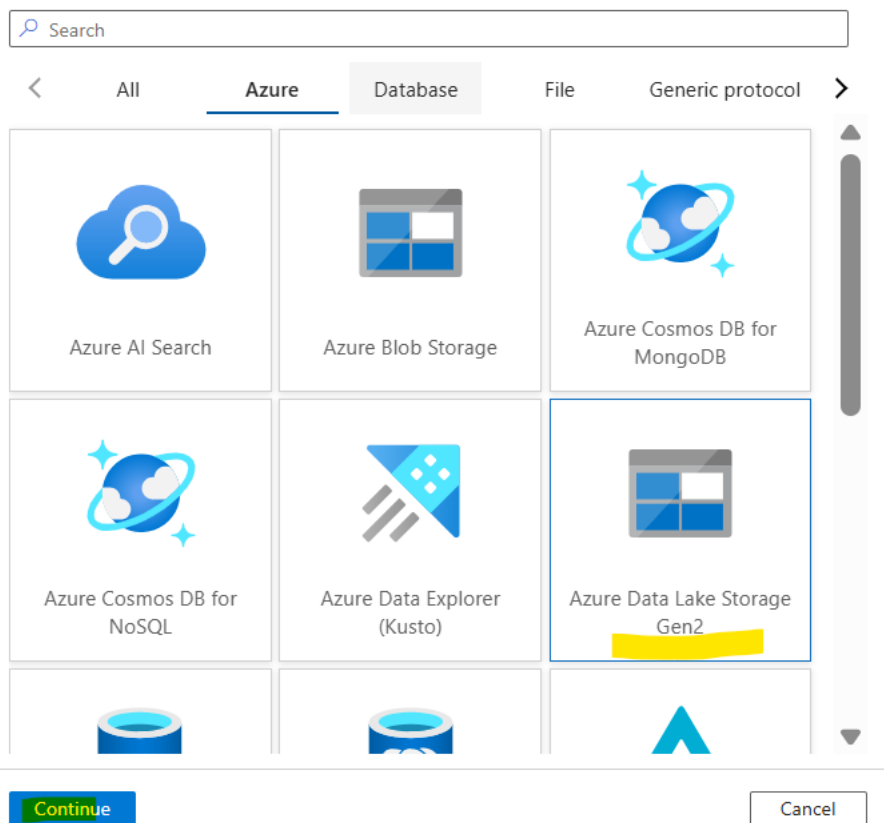


Select the storage account(adls) and the file type

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store



Select format

Choose the format type of your data

 Avro	 Binary	 DelimitedText
 Excel	 Iceberg	 JSON
 ORC	 Parquet	 XML

Continue

Back

Cancel

Create a new linked service to copy data from blob---give subscription---storage account---test connection

New linked service

 Azure Data Lake Storage Gen2 [Learn more](#)

Connect via integration runtime * ⓘ

 AutoResolveIntegrationRuntime

Authentication type

Account key

Account selection method ⓘ

☒ From Azure subscription ☐ Enter manually

Azure subscription ⓘ

NK (5bd8f1cd-03ad-48fa-8410-6e58274fc11f)

Storage account name *

pipeline3adlsa



Test connection ⓘ

☒ To linked service ☐ To file path

Annotations

+ New

> Parameters

> Advanced ⓘ

Create

Cancel

 Connection successful

 Test connection

Provide the file path

Set properties

Name

Training_data

Linked service *

adls2adf

File path

practice

/ dataset

/ Training1 (1).csv

First row as header ☒

Import schema

☒ From connection/store ☐ From sample file ☐ None

OK

Back

Cancel

Preview the data, to see the data is loaded correctly or not

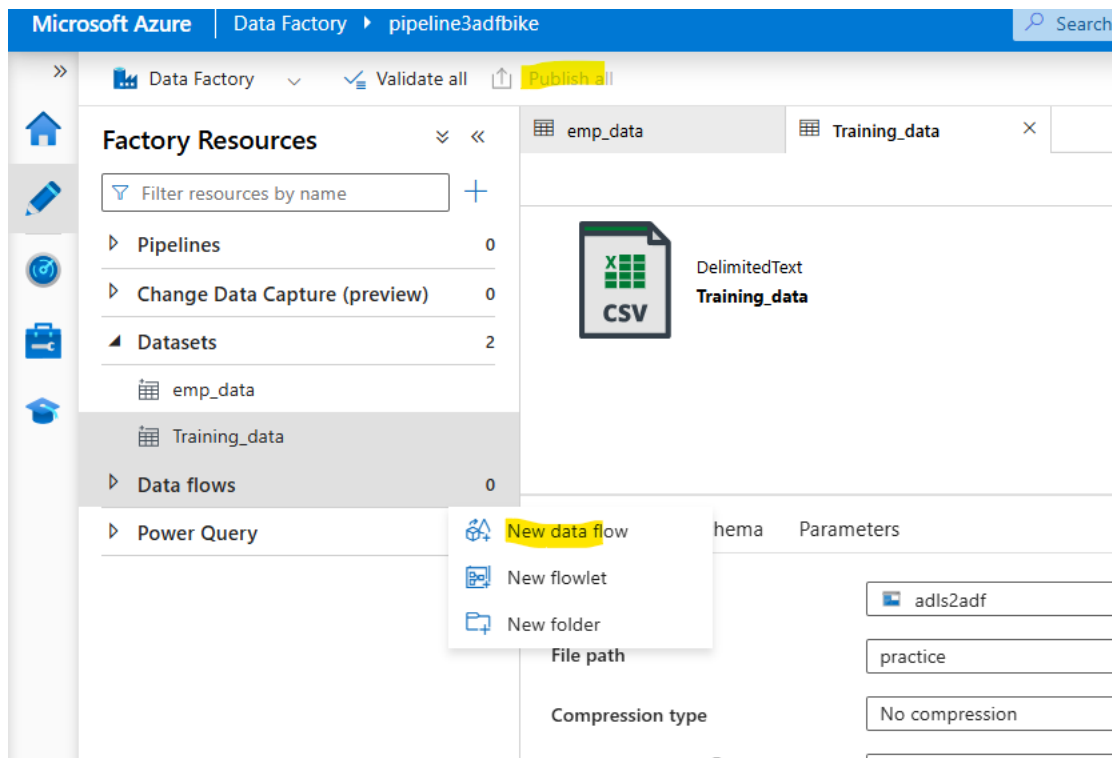
The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'Pipelines' (0), 'Change Data Capture (preview)' (0), 'Datasets' (2), 'Data flows' (0), and 'Power Query' (0). The 'Datasets' section is expanded, showing 'emp_data' and 'Training_data'. The 'Training_data' dataset is selected. The main pane displays the 'Preview data' window for the 'Training1 (1).csv' object. The 'Linked service' is 'adls2adf'. The preview shows a table with 7 rows and 7 columns: Start Date, Course Code, Course Name, Employee_ID, Cost, and Supplier. The data is as follows:

	Start Date	Course Code	Course Name	Employee_ID	Cost	Supplier
1	18-09-2015	1	Communication Workshop	1	1300	Communication Experts
2	18-09-2015	1	Communication Workshop	4	1300	Communication Experts
3	18-09-2015	1	Communication Workshop	25	1300	Communication Experts
4	18-09-2015	1	Communication Workshop	33	1300	Communication Experts
5	18-09-2015	1	Communication Workshop	44	1300	Communication Experts
6	18-09-2015	1	Communication Workshop	23	1300	Communication Experts
7	18-09-2015	1	Communication Workshop	6	1300	Communication Experts

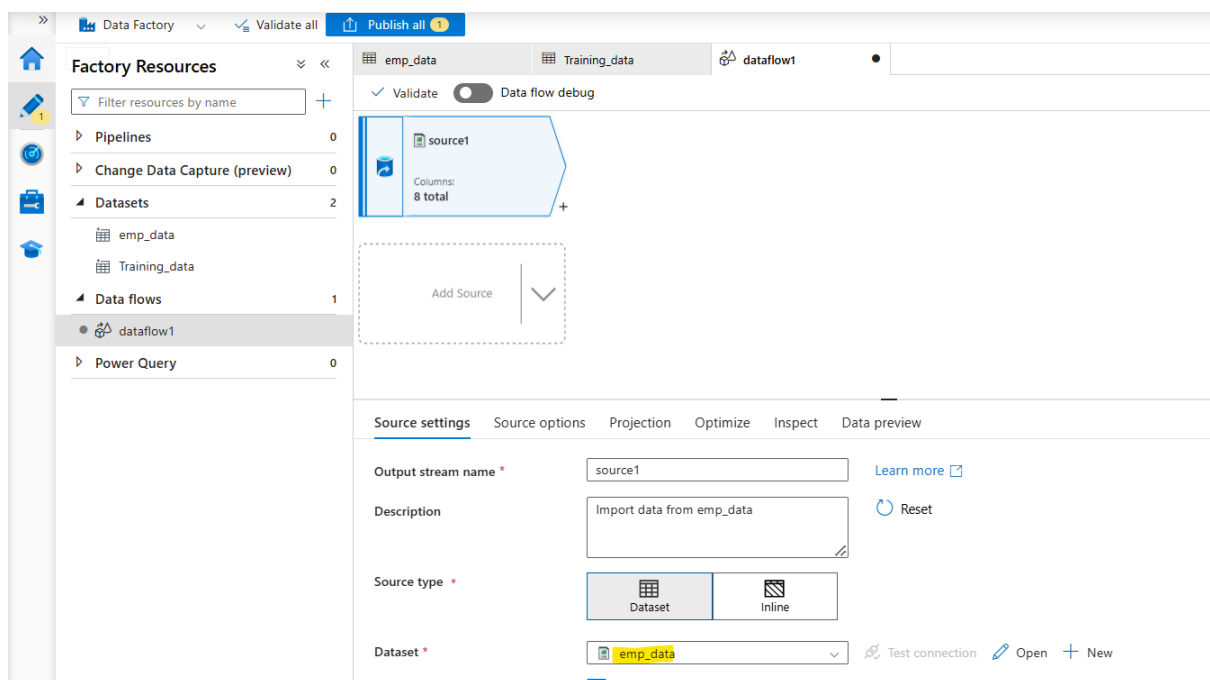
On the right side of the preview window, there are tabs for 'Properties', 'Schema', and 'Annc'. The 'Properties' tab is active, showing details for the dataset.

*****Publish all the changes to adf(to avoid data loss)

Create data flow for transformations:



Now we have two datasets hence we need to create two sources and join the data



Turn on the data flow debug mode to preview the data

» Data Factory Validate all Publish all 1

Factory Resources

- Pipelines 0
- Change Data Capture (preview) 0
- Datasets 2
 - emp_data
 - Training_data
- Data flows 1
 - dataflow1
- Power Query 0

emp_data Training_data dataflow1

Validate Data flow debug

source1
Columns: 8 total

Add Source

Source settings Source options Projection Optimize Inspect Data preview

Output stream name * source1 [Learn more](#)

Description Import data from emp_data [Reset](#)

Source type * Dataset Inline

Dataset * emp_data [Test connection](#) [Open](#) [New](#)

Turn on data flow debug

Integration runtime

AutoResolveIntegrationRuntime

AutoResolveIntegrationRuntime

Region AutoResolve

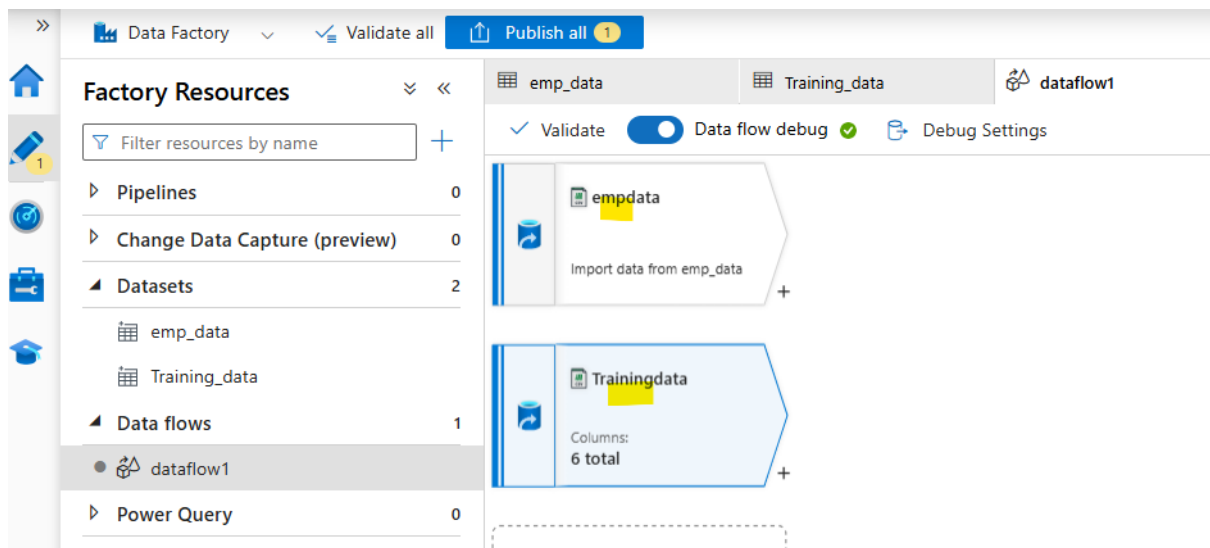
Compute size Small

Debug time to live ⓘ

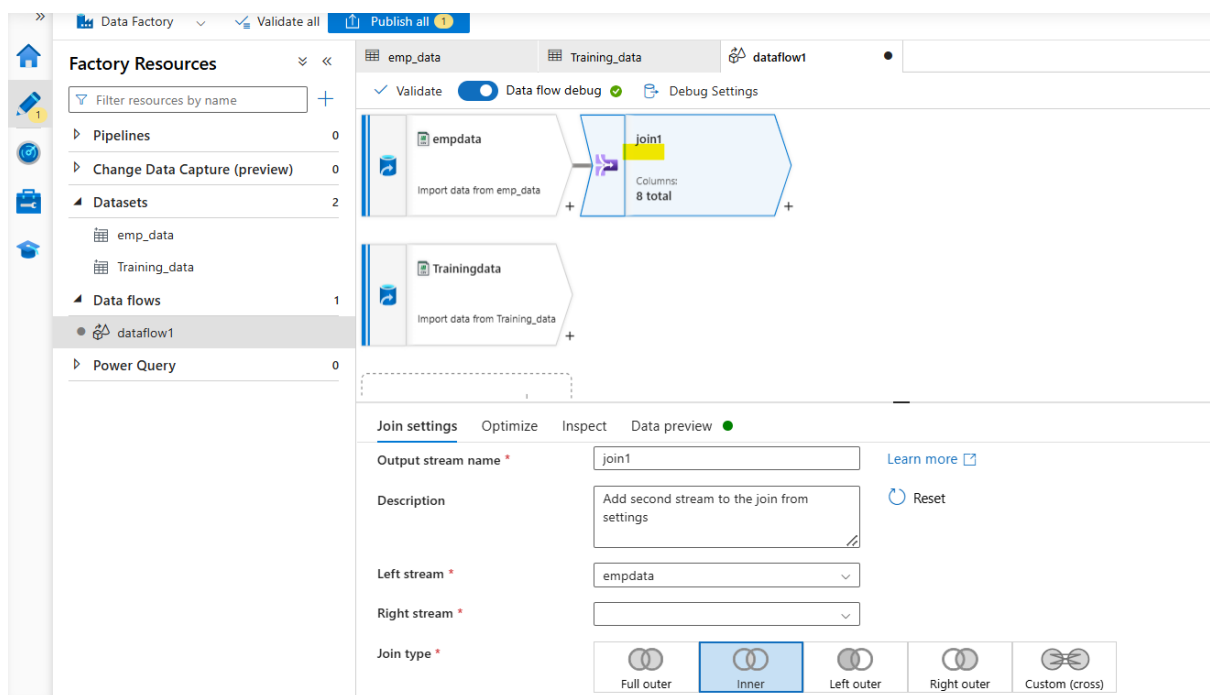
1 hour

OK Cancel

Now we have 2 sources –1 is connected with emp data-another is connected with training data



We need to join both the tables----click +---select (join)



Once we select the join we need to provide the join type, and the columns need to join on both tables

Join settings Optimize Inspect Data preview ●

← Previous Next →

Output stream name * [Learn more](#)

Description [Reset](#)

Left stream *

Right stream *

Join type *

Full outer
 Inner
 Left outer
 Right outer
 Custom (cross)

Use fuzzy matching ☐

Join conditions *

Left: empdata's column Right: Trainingdata's column

[Open expression builder](#)

Now we need to see the columns, remove the columns which are not necessary and rename the column names if needed, can change data type

Successfully changed the required columns

Select settings Optimize Inspect Data preview ●

← Previous Next →

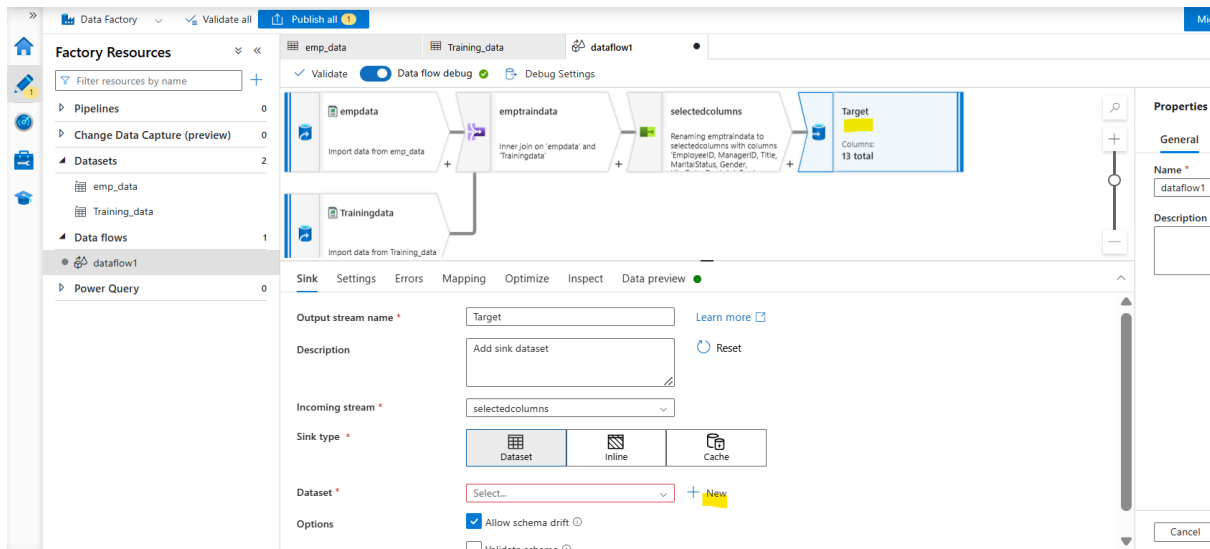
☒ Skip duplicate output columns

Input columns *

☐ Auto mapping [Reset](#) [+ Add mapping](#) [🗑️ Delete](#) 13 mappings: 1 column(s) from the inputs left unmapped

☐	emptraindata's column		Name as		
☐	abc EmployeeID	→	EmployeeID		+ 🗑️
☐	abc ManagerID	→	ManagerID		+ 🗑️
☐	abc Title	→	Title		+ 🗑️
☐	abc MaritalStatus	→	MaritalStatus		+ 🗑️
☐	abc Gender	→	Gender		+ 🗑️
☐	abc HireDate	→	HireDate		+ 🗑️
☐	abc Dept	→	Dept		+ 🗑️
☐	abc Job Grade	→	JobGrade		+ 🗑️
☐	abc Start Date	→	StartDate		+ 🗑️
☐	abc Course Code	→	CourseCode		+ 🗑️
☐	abc Course Name	→	CourseName		+ 🗑️
☐	abc Cost	→	Cost		+ 🗑️
☐	abc Supplier	→	Supplier		+ 🗑️

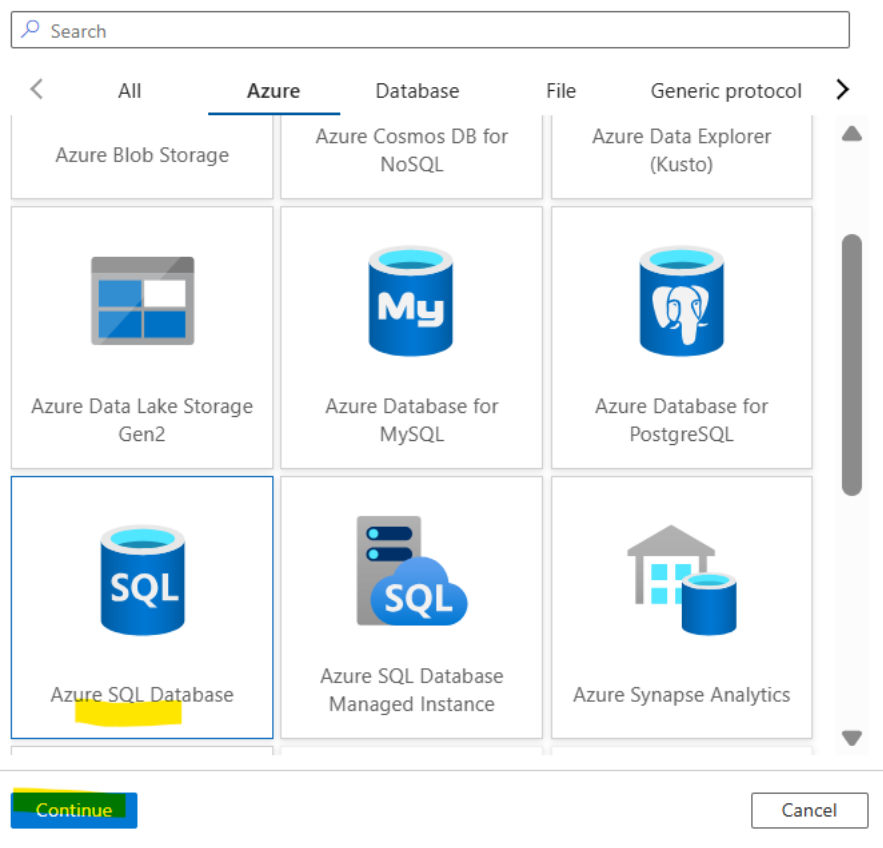
Once the data is cleaned we need to sink the data(copy to target)



New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store



Create a linked service to copy the data from adf to target(azure sql db)---give subscription name,server name,db name,and sql authentication---testconnection---create---then create a new table

New linked service

 Azure SQL Database [Learn more](#)

☒ 2.0 (Recommended) ☐ 1.0

 Import from connection string

Account selection method ⓘ

☒ From Azure subscription ☐ Enter manually

Azure subscription

NK (5bd8f1cd-03ad-48fa-8410-6e58274fc11f) ▾

Server name *

pipeline3server ▾ 

Database name *

pipeline3db ▾ 

Authentication type *

SQL authentication ▾

User name *

adminnk

Password

[Azure Key Vault](#)

Password *

Always encrypted ⓘ ☐

Create

Cancel

✔ Connection successful

 Test connection

Set properties

Name

emptrain_data

Linked service *

adf2azsqldb

☐ Select from existing table ☒ New table

Schema and table name

Schema name

EmpTraining_TB

> Advanced

OK

Back

Cancel

Publish all the changes in the adf before we create a pipeline

The screenshot displays the Azure Data Factory (ADF) interface. On the left, the 'Factory Resources' pane shows a tree view with 'Pipelines' (0 items), 'Change Data Capture (preview)' (0 items), 'Datasets' (3 items: emp_data, emptrain_data, Training_data), and 'Data flows' (1 item: dataflow1). The main canvas shows a data flow named 'dataflow1' with the following steps: 'empdata' (Import data from emp_data), 'emptraindata' (Import data from Training_data), 'selectedcolumns' (Inner join on 'empdata' and 'Trainingdata', then Renaming emptraindata to selectedcolumns with columns: EmployeeID, ManagerID, Title, MaritalStatus, Gender, ...), and 'Target' (Columns: 13 total). Below the canvas, the 'Sink' settings for the 'Target' step are configured: 'Output stream name' is 'Target', 'Description' is 'Export data to emptrain_data', 'Incoming stream' is 'selectedcolumns', 'Sink type' is 'Dataset', 'Dataset' is 'emptrain_data', and 'Options' include 'Allow schema drift' checked. On the right, the 'Properties' pane shows the 'General' tab with 'Name' as 'dataflow1' and an empty 'Description' field. A 'Publishing' notification at the top right indicates 'Deploying changes to the factory'.

TO CREATE PIPELINE:

» Data Factory Validate all Publish all

Factory Resources

Filter resources by name

- Pipelines 0
 - Change Data Capture (preview)
- Datasets
 - emp_data
 - emptrain_data
 - Training_data
- Data flows 1
 - dataflow1
- Power Query 0

emp_data Training_data

Validate Data flow debug Debug Setting

Reference: emptraindata

Inner join on 'empdata' and 'Trainingdata'

Trainingdata

Import data from Training_data

Sink Settings Errors Mapping Optimize Inspo

Output stream name * Target

» Data Factory Validate all Publish all

Factory Resources

Filter resources by name

- Pipelines 1
 - pipeline1
- Change Data Capture (preview) 0
- Datasets 3
 - emp_data
 - emptrain_data
 - Training_data
- Data flows 1
 - dataflow1
- Power Query 0

Activities

Search activities

- Move and transform
 - Copy data
 - Data flow
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

Data flow

Data flow1

General Settings Parameters User properties

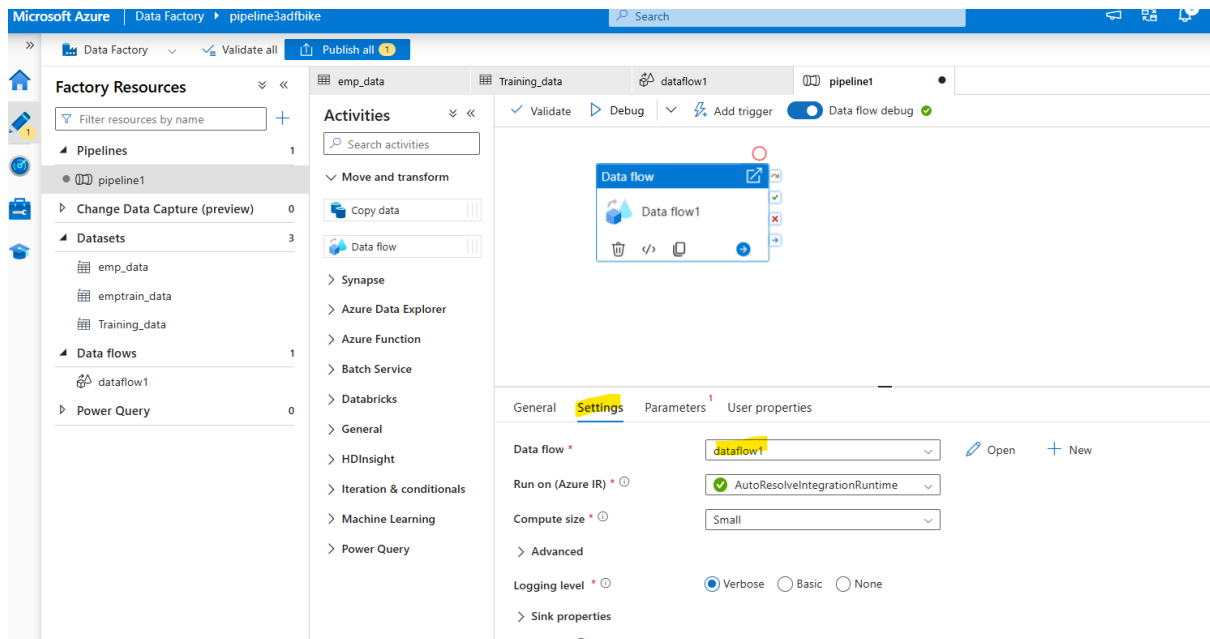
Name * Data flow1 [Learn more](#)

Description

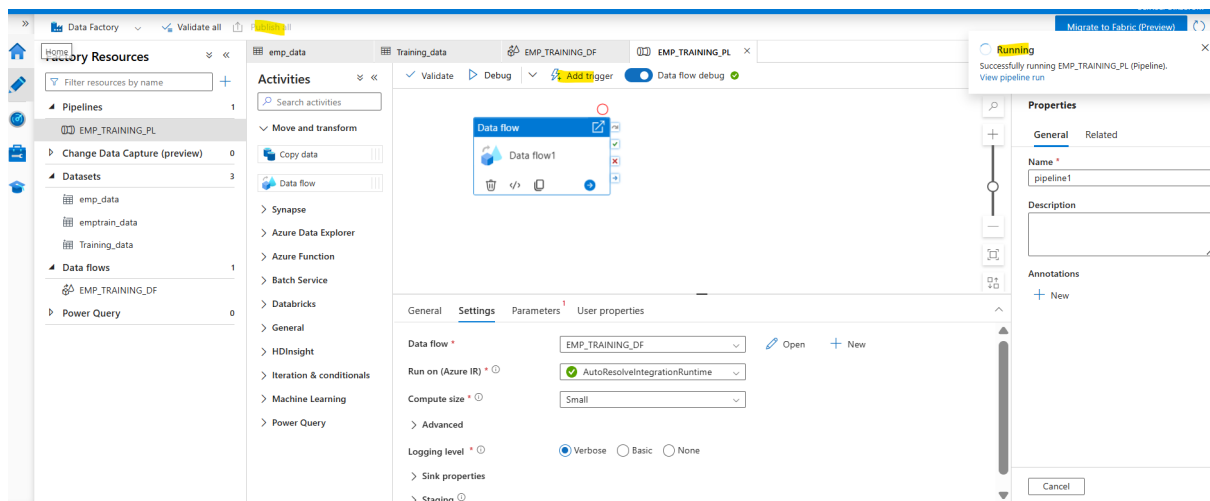
Activity state ☒ Activated ☐ Deactivated

Timeout 0.12:00:00

Retry 0



Publish all the changes ---add trigger---triggernow---run the pipeline



Successfully pipeline is executed

Microsoft Azure | Data Factory | pipeline3adbike

All pipeline runs > EMP_TRAINING_PL - Activity runs

Runs

Pipeline runs

Trigger runs

Change Data Capture (previ...

Runtimes & sessions

Integration runtimes

Data flow debug

Notifications

Alerts & metrics

Data flow

Data flow1

Activity runs

Pipeline run ID d0b75128-3279-4166-aa2e-7235d11b1a11

All status

Showing 1 - 1 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity run ID
Data flow1	Succeeded	Data flow	6/24/2026, 2:59:25 PM	4m 21s	AutoResolveIntegrationRuntime (East US)		55916e2b-f93c-4

Check in the target(Azure SQL DB):

Home > Microsoft.SqlDatabase.newDatabaseNewServer_3b9f0fc844644a4aa6b3a | Overview > pipeline3db (pipeline3server/pipeline3db)

pipeline3db (pipeline3server/pipeline3db) | Query editor (preview)

SQL database

Search

New query

Templates

Open in

Feedback

Sign out

Classic experience

Overview

Activity log

Tags

Diagnose and solve problems

Query editor (preview)

Mirror database in Fabric (preview)

Resource visualizer

Settings

Data management

Integrations

Power Platform

Security

Intelligent performance

Monitoring

Automation

Help

Explorer

Search

pipeline3db

dbo

Tables

Emp_Training_TB

Views

Stored Procedures

Functions

Queries

SQL query 1

SQL query 1

Run

Save as view

1 SELECT TOP (1000) * FROM [dbo].[Emp_Training_TB]

Messages

Results

Search

	ARC Employee...	ARC ManagerID	ARC Title	ARC MaritalSt...	ARC Gender	ARC HireDate	ARC Dept	ARC JobGrade	ARC StartDate	ARC Course...	ARC CourseN...
1	1	16	Gustavo Acho...	M	M	02-02-2013 0...	Sales	Admin	18-09-2015	1	Communicati...
2	4	3	Humberto Ac...	S	M	10-07-2014 0...	Logistics	Admin	18-09-2015	1	Communicati...
3	25	21	Phyllis Allen	S	M	09-07-2015 0...	Production	Operations	18-09-2015	1	Communicati...
4	33	135	Maxwell Amla...	S	M	13-07-2015 0...	Sales	Operations	18-09-2015	1	Communicati...
5	44	148	John Ault	S	M	18-07-2015 0...	Sales	Operations	18-09-2015	1	Communicati...
6	23	197	Michelle Alex...	S	M	08-07-2015 0...	Human Resou...	Management	18-09-2015	1	Communicati...
7	6	109	Frances Adams	S	M	25-07-2014 0...	Sales	Management	18-09-2015	1	Communicati...
8	2	6	Catherine Abel	S	M	31-08-2013 0...	Sales	Management	22-01-2017	2	Communicati...
9	5	263	Pilar Ackeman	M	M	16-07-2014 0...	Human Resou...	Admin	22-01-2017	2	Communicati...

Succeeded (0 sec 665 ms)

Columns: 13 Rows: 419

Sucessfully copied the transformed data to the target

